N-R-002146(Army-GL)
September 24, 1982

INTERIM FEDERAL SPECIFICATION

RICE, SPANISH, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers spanish rice thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 8940

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Military Specification:

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 100-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Institutional Meat Purchase Specifications for Cured, Cured and Smoked, and Full-Cooked Pork Products Approved by USDA

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Food Safety and Inspection Service, U.S. Department of Agriculture, Meat and Poultry Inspection Service, Washington, DC 20250.)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products

United States Standards for Grades of Onions (Other Than Bermuda - Granex Grano and Creole Types)

United States Standards for Grades of Canned Mushrooms

United States Standards for Sweet Peppers

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(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

United States Standards for Milled Rice

(Application for copies should be addressed to the Federal Grain Inspection Service, Resource Management Division, 1400 Independence Avenue, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.4).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Rice. Rice shall be a 50/50 mixture of U.S. No. 1 Grade Long Grain Milled Rice and Medium Grain Milled Rice of the United States Standards for Milled Rice.

3.2.2 Sliced bacon. Sliced bacon shall comply with item No. 541 - Bacon, Sliced (Cured and Smoked) Ends and Pieces of the Institutional Meat Purchase Specification approved by USDA.

3.2.3 Onions. Onions shall be yellow, dry, and firm. Onions shall not be Bermuda-Granex and Creole types. Onions shall be Commercial Grade of the United States Standards for Grades of Onions (Other Than Bermuda-Granex-Grano and Creole Types).

3.2.4 Green peppers. Green peppers shall be firm, crisp, and free from rot or mold. The green peppers shall be US No. 2 Grade or higher of The United States Standards for Sweet Peppers.

3.2.5 Sweet red peppers. Sweet red peppers shall be firm and free from rot or mold. The sweet red peppers shall be US No. 2 Red Grade or higher of the United States Standards for Sweet Peppers.

3.2.6 Mushrooms. Mushrooms shall be any style white U.S. Grade A of the United States Standards for Grades of Canned Mushrooms.

3.2.7 Tomatoes. Tomatoes shall be canned Italian style, pear tomatoes in thick tomato sauce.

3.2.8 Cumin. Cumin shall be the ground, true aromatic vegetable substance derived from Cuminum cyminum L. from which no volatile or other flavoring constituent has been removed and shall be free from artificial coloring, exhausted spices, adulterants and impurities.

3.2.9 Garlic powder. Garlic shall be dehydrated garlic powder that is free flowing, uniformly light in color, pungent, and practically free from outer skin and scorched particles. Garlic powder shall be free from husk, root base, stem, soil, and mold.

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3.2.10 Monosodium glutamate. Monosodium glutamate shall be odorless, free flowing and may have either a slightly sweet or slightly salty flavor. Monosodium glutamate crystals shall be white and translucent.

3.2.11 Salt. Salt shall be free from extraneous material. Commonly used recrystallized or vacuum pan granulated salt, with or without anti-caking agent is acceptable. Iodized salt shall not be used.

3.2.12 Starch, modified. Modified starch shall be a white, odorless, finely pulverized, modified waxy maize food starch for use in canned foods. The starch shall be bland with essentially no cereal or starchy taste and shall comply with the bacteriological requirements of MIL-STD-900.

3.2.13 Sweetening ingredients. Sweetening ingredients shall be white, refined, granulated cane or beet sugar, liquid sugar, invert sugar or any combination of invert sugar with corn syrup or corn syrup solids or dextrose. Sweetening ingredients shall conform to MIL-STD-900.

3.2.14 Turmeric. Turmeric shall be the dark yellow, aromatic, vegetable powder from Curcuma longa L. No volatile or other flavoring constituent shall be removed. Turmeric shall be free from artificial coloring, adulterants and impurities.

3.2.15 White pepper. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. Volatile oil content shall not be less than 1.0 mL per 100 grams of pepper.

3.3 Rice preparation. Rice shall be blanched sufficiently to meet the finished product requirements of 3.8.

3.4 Sauce preparation. The sauce shall be prepared from the following ingredients:

Ingredient	Percent by weight
Bacon, sliced, raw	9.50
Onions, chopped	9.70
Green peppers, chopped	9.40
Sweet red peppers, chopped	4.20
Mushrooms, chopped	11.10
Tomatoes	40.20
Cumin	0.01
Garlic powder	0.03
Monosodium glutamate	0.03
Salt	0.50
Starch, modified	0.10
Sugar	2.50
Turmeric	0.01
White pepper	0.03
Water	12.69
	100.00

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The sauce shall be prepared as follows:

a. The bacon, cut transversely in approximately 3/8-inch wide strips, shall be sauteed lightly.

b. The onions, red peppers, green peppers and mushrooms shall be chopped into approximately 3/8-inch diameter pieces, added to the drippings, and sauteed lightly.

c. The remaining ingredients shall be added. The mixture shall be heated to 212°F and then simmered for 5 minutes.

d. The sauce shall be used within a 4-hour period after preparation and held at 150 ± 10°F during this period.

3.5 Spanish rice preparation. The blanched rice and the prepared sauce shall be thoroughly mixed in the proportion by weight of 30 percent rice to 70 percent sauce.

3.6 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the following requirements:

a. The temperature of the product at time of filling shall be 150 ± 10°F.
b. The average net weight shall be 106 ounces minimum.
c. No individual can shall contain less than 104 ounces of product.
d. The can shall be hermetically sealed immediately after filling.

3.7 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 250°F until a sterilization value (F_o) of not less than 6.0 has been achieved.

3.8 Finished product requirements. The finished product shall have the following characteristics. The Spanish rice shall be flavorful and shall have no off-odors or off-flavors such as scorched, moldy, rancid, sour or stale. The Spanish rice shall be colorful with individual components uniformly distributed and not materially darkened, scorched, or oxidized. Vegetables shall be distinct dices and not soft or mushy. The rice granules shall be firm and not hard, mushy or pasty. The salt content of the finished product shall be not less than 0.5 nor more than 1.1 percent. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.9 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

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3.10 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting it to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected for palatability, for overall appearance, and for the defects listed in table I and it shall be tested for the characteristic indicated in 4.7.3. Palatability failure, appearance failure, the presence of any defect, or any test failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade Certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.6 In-process examination. In-process examination shall be performed to determine conformance to the ingredient preparation, product processing, can interior coating, and sealing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation; certificates of conformance; or other valid documents shall be cause for rejection of the involved product.

4.7 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

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4.7.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected. The results shall be reported to the nearest 0.1 ounce.

4.7.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated at least 15 minutes in boiling water, opened, and inspected for the defects listed in table I.

a. If no defects are found, the lot shall be acceptable.

b. If two or more defects are found, the lot shall be rejected.

c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, and inspected for the defects listed in table I. If any defect is found in the second sample, the lot shall be rejected.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	Individual components not uniformly distributed.
152	Color not characteristic of product (materially darkened, scorched, or oxidized).
153	Texture of rice hard, mushy, or pasty.
154	Vegetables not distinct units.
155	Texture of vegetables soft and mushy.
156	Flavor not characteristic of Spanish rice.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by the US Department of Agriculture inspection agent).

4.7.3 Salt content testing. Nine tray packs shall be selected from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by the inspector as a reserve sample

The three tray packs for laboratory analysis shall be individually tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the sample shall be as follows. The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if the salt content in any individual can is less than 0.5 or more than 1.1 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve samples are analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.8 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers.

4.9 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used. Level C shipping containers shall be examined only for the marking defects heretofore specified.

4.10 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.6 and 3.7. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2.3.1 Alternate level C packing. When specified (see 6.1), four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade

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275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension and be fabricated of class domestic fiberboard.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to Navy, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each package shall be labeled to comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Paper labels are not permitted. The name of the product or abbreviation thereof and date of pack (day, month, year) shall be embossed or lithographed on the end (lid) of the can. Cans shall have the following heating directions and yield:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 20 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 20 minutes until contents reach an internal temperature of 165°F.
WARNING. Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.4).

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(c) Provisions for approved preproduction sample (see 3.8 and 6.4).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(f) When alternate (special requirement) level C packing is required (see 5.2.3.1).
(g) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

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